

No Code App Builder

*

1st Rajni Tiwari

CMR Institute of Technology
Bengaluru, Karnataka, India
rajni.t@cmrit.ac.in

2nd Muskan Shah

CMR Institute of Technology
Bengaluru, Karnataka, India
mush21cs@cmrit.ac.in

3rd Neha Baiju

CMR Institute of Technology
Bengaluru, Karnataka, India
neba21cs@cmrit.ac.in

4th Raghav Saxena

CMR Institute of Technology
Bengaluru, Karnataka, India
rasa21is@cmrit.ac.in

Abstract—This open-source drag-and-drop website builder is intended to simplify indicating that the users need to drag and drop the widgets onto a canvas to create their expected pages other than code for a website. The drag and drop interface give the user a hands-on experience in real-time adding, editing, removing, and organizing web components while free enabling creativity and productivity. It is built using HTML, CSS, and JavaScript; hence it is quite versatile. From the end-user point of view, the No Code Web App builder offers an easy-to-use design for better adoption while at the same time keeps the flexibility required to have advanced customizations. The paper focuses on making the process of creating websites faster, easier, and more accessible to larger audiences, hence closing the gap between professional web development tools and more everyday usability for individuals and firms alike.

Index Terms—Open source, Drag-and-drop, End-user design, Advanced customizations, No-code.

I. INTRODUCTION

In the modern digital world, web development stands for communication, business, and innovations. With the advancing technological wonder, there has also been a lot of demand for dynamic and easily used websites. Thus, various tools have to fill the existing gap between professional developers and an individual with little technical knowledge. No Code Web-App Builder is one such platform that has developed an open-source drag-and-drop website builder app to satisfy this need. With intuitive design and highly functional utility, the No Code Web-App Builder makes the process of creating, customizing, and deploying websites quick for everyone, irrespective of their technical skills.

The design of No Code Web-App Builder is based on modern web technologies such as HTML, CSS, and JavaScript, which exemplify the compatibility and performance that should exist over various environments. The most striking feature is the real-time editing interface that allows users to drag-and-drop, resize, style, and organize their layout. No Code Web-App Builder can allow non-programmers to build websites without losing flexibility or customization available to seasoned developer audiences as compared to traditional creating approaches, which very often took up a lot of coding at certain points during development.

No Code Web-App Builder aims to efficiency and creativity. It does not repeat coding tasks for users and instead con-

centrates on design and function. The modular form of this tool provides for a wide range from text blocks to images, forms, and interactive elements that can be integrated into any paper. Besides, the tool has real-time previews, which help users visualize changes in a split second and to iterate designs fruitfully. All these aspects cut down development time but stand tall against quality and precision standards. The truly remarkable thing about No Code Web-App Builder is that it is open source. It gives accessibility to use of the source code to be able to add features, fix bugs, or create new components. By doing this, the platform encourages raising community level enhancements and shared development toward the paper. This is introspective of creative activities and gives a firm assurance that No Code Web-App Builder will keep face with a changing world in web development trends and techniques.

In conclusion, No Code Web-App Builder represents a paradigm shift in web development, combining the accessibility of drag-and-drop tools with the power and flexibility of traditional coding. Its open-source nature, user-friendly interface, and focus on efficiency make it a valuable asset for both novice users and experienced developers. As digital transformation continues to accelerate, tools like No Code Web-App Builder will remain instrumental in empowering individuals and organizations to thrive in the online world. This paper report delves into the development, features, and impact of No Code Web-App Builder, highlighting its contributions to simplifying and democratizing web development.

II. LITERATURE REVIEW

A survey of the available literature illustrates a few of the techniques that low-code platforms and no-code platforms have adopted, primarily their use, strengths, and limitations in different domains. The research studies various methodologies, namely visual development environments, dynamic integrations, or machine-learning-driven tools, offering insights into the potentialities and the challenges. The survey indicates simplified application development using these platforms and reflects on voids that need to be filled for their successful application in various environments. Drag and drop tools are basically one of the most played-out processes with regard to no-code platforms that enable users, without really indulging in deep technical learning, to create visually designed applications. These transform usually complex coding tasks

into simple, intuitive actions and enable users, even the non-technical, to create working applications in no time at all. Probably the biggest selling point for these platforms particularly bothers small businesses, startups, and even individuals who want to have something web or mobile-focused in a way but don't have much in terms of programming skills. However, while drag- and-drop tools offer methodologies for very fast development, their capabilities often extend only to the simplest of applications; they predominantly fail in dealing with more advanced workflows or leveraging sophisticated features that might really require other kinds of sophisticated tools or custom code. Moreover, these platforms are not very flexible either, since they do not support elaborate customization or scalable solutions for big enterprises.

The paper [1][15] surveys for low-power devices and real-time applications. The paper explores low-code platforms to make software and ML development more accessible, allowing non-experts to create applications with minimal coding. It reviews platform capabilities and limitations, analyzes Stack Overflow discussions to understand practitioner challenges, and develops a case study with AutoGeoML—a toolkit for geothermal ML applications. Key findings include issues with customization, deployment, and resource demands. The paper provides recommendations for improving usability, documentation, and domain-specific customization in low-code platforms.

The paper [2][10] explores the intersection between low-code development platforms (LCDPs) and model-driven engineering (MDE), two approaches that aim to simplify software development but differ in execution. LCDPs reduce coding through visual interfaces, targeting non-programmers and accelerating deployment, while MDE uses detailed models to automate complex development tasks, often needing more expertise. The paper suggests each approach has unique strengths and highlights ways they can cross-pollinate—for example, by bringing MDE's rigorous model-based methods into LCDPs to enhance flexibility and adopting LCDPs' user-friendly, cloud-based environments in MDE.

The paper [3] discusses the advantages of no-code platforms, specifically Logotec App Studio, in making software development accessible and affordable for non-programmers and small businesses. Unlike traditional programming, Logotec App Studio enables users to build applications through drag-and-drop features, significantly reducing costs and development time. While this approach empowers those without coding skills, it may not suit highly complex papers requiring extensive customization.

The paper [4][12] introduces low-code development is transforming software creation by leveraging visual tools and automation to minimize traditional coding efforts. This approach enables both IT professionals and non-technical business users to collaborate effectively, accelerating the development cycle and reducing costs. With Agile and RAD methodologies, low-code platforms promote flexibility and quick iterations. However, challenges like limited customization and potential vendor lock-in must be addressed. The benefits, including

faster delivery, cost savings, and enhanced collaboration, make it a popular choice for organizations aiming to respond swiftly to market demands while empowering business teams to develop their own solutions.

The paper [5][11] proposes a low-code/no-code platform to meet the rising demand for mobile applications, especially within enterprises. As IT departments struggle to keep up, platforms like the Smart Maker Authoring Tool allow non-developers to design and deploy apps using visual interfaces and automated tools. By enabling faster and easier app development, businesses can reduce costs and accelerate their digital initiatives. However, while the in advanced application development.

The paper [6][13][15] examines how LCDPs empower non-IT employees to contribute to digital innovation, especially in scenarios where rapid prototyping and user involvement are critical. By lowering the technical barriers, LCDPs enable "citizen developers" to create tailored solutions faster and more affordably. However, while these platforms democratize software development, they also introduce challenges in usability and customization, suggesting a need for balanced implementation strategies to avoid quality trade-offs in user experience and performance.

The paper [7] evaluates various NoCode tools like Sound-Draw.io for audio generation, Wix for website building, and Figma for collaborative design. These platforms empower non-coders to quickly create and manage digital content, thus optimizing workflows in game design. While these tools significantly reduce development time and lower entry barriers, they may have limitations in customization for more complex papers. Overall, they enable creators to focus on design rather than technical implementation, making them valuable assets in modern digital content creation.

The paper [8] discusses an innovative citizen science app platform that enables non-experts to collect environmental data using mobile devices. The system simplifies the creation of mobile applications integrated with machine learning (ML) for image detection tasks. Users can label images, train ML models, and create apps without coding. With a focus on client-side processing, the platform allows real-time object detection without network dependency, supporting diverse data collection tasks.

The paper [9] presents Pathverse, a no-code platform designed to help researchers develop mobile health (mHealth) intervention apps tailored for behavior change without needing coding skills. Guided by behavior change theories and the behavior change technique taxonomy, Pathverse offers customizable features like multimedia content, self-monitoring tools, and goal tracking, aimed at supporting health interventions in a user-friendly format. The platform utilizes a participatory development approach, involving user feedback to refine functionalities continuously. Although it simplifies app creation, it currently lacks some customization and scalability, which future updates aim to address to enhance user experience and adaptability.

III. PROPOSED ARCHITECTURE

Our proposed system architecture design allows the users to view and manage the templates and widgets in the drag-and-drop format. It emphasizes a complete separation of user interaction with the frontend and database management and all other processing in the backend.

a. System Architecture: The proposed model of No Code Web App Builder would bring up the core feature by employing an intelligent, modular design adaptable to user needs. A system based on component architecture wherein all UI elements are defined as reusable modules makes the creation of pages easier to build and customize. By using drag-and-drop mechanisms, snapping to grid and dynamic binding of properties, it assures users that they will be able to create responsive and visually consistent web pages without learning too much about programming. Moreover, a server-side integration layer for integrating real-time collaboration and content management has been added to the model. Saving and fetching content with PHP scripts, JSON- based configurations for templates, and deploying the tool in different environments using Docker containers. Moreover, the task automation pipeline is supplemented with Gulp, allowing asset management, SCSS compilation, and development workflows to be more effective.



Fig. 1. System Architecture

b. Data Flow: The users interact with the Drag-and-Drop Builder for the creation and modification of templates using the dragging and dropping of widgets. This interaction is handled, updated in the visual representation, through the Core JavaScript Library. When a user saves their design, the Export/Save Module serializes the design data. This serialized data is sent to the backend and the Template Storage. When the need arises for a template to be rendered, Template Engine retrieves the template data from the Template Storage processes it, and generates an output. The backend also has its separate management of widgets and user performance.

c. Minimum Hardware and Software Requirements: The no code app builder requires specific hardware and software configurations for optimal performance. On the server hardware side, a machine with a minimum of 16 GB RAM, storage of 100 GB SSD, operating system: Linux (Ubuntu 20.04 or higher) or Windows Server 2019. On the user hardware side, a machine with a minimum of 8 GB RAM, storage of 20 GB, operating system: Windows 10, macOS 10.15, or higher.

IV. IMPLEMENTATION

The No Code App Builder is a revolutionary platform designed to simplify website creation for users of all skill levels by embracing HTML, CSS, and JavaScript as its foundation. Its user-centric approach empowers both beginners and seasoned developers to design and build professional-grade websites effortlessly. One of the platform's standout features is its intuitive drag-and-drop interface, which allows users to arrange components seamlessly on the canvas, eliminating the complexities typically associated with web development. The live editing system enhances the design experience by directly linking input fields and property panels to the DOM, enabling immediate inline editing and providing real-time feedback as changes are made. A key aspect of the platform is its focus on reusability; components such as blocks, elements, and sections are thoughtfully designed to minimize redundant customization efforts while maintaining flexibility for creative freedom.

Beyond visual design, the platform integrates robust backend capabilities, ensuring efficient management of templates and enabling the creation of plugin-extensible designs. These features work in harmony to produce clean, export-ready HTML, CSS, and JavaScript files, ensuring compatibility and ease of deployment across various hosting environments. The server backend is optimized to handle saving and loading papers, enhanced further by dynamic configuration options that offer unparalleled flexibility to adapt to different paper requirements. Additionally, the open-source nature of the paper fosters an active community of contributors and users, encouraging continuous improvements, new features, and widespread adoption. This collaborative model not only accelerates the evolution of the platform but also ensures it remains accessible and relevant to a diverse audience.

Overall, the No Code Web-App Builder bridges the gap between simplicity and functionality, democratizing website creation by providing an inclusive, efficient, and powerful tool for individuals, businesses, and developers alike. By combining modern design principles, robust backend support, and an open-source philosophy, the platform redefines how websites are built, enabling faster workflows, better collaboration, and a more accessible web development process for all.

It is begun by setting up the development environment, ensuring all necessary dependencies are configured, and gaining a comprehensive understanding of the core paper files and their functionality. Customize the user interface by making modifications to the HTML structure, updating CSS styles to match the desired branding and design, and refining JavaScript to enhance functionality and responsiveness. Expand the component library by defining new structures, specifying editable options, and integrating drag-and-drop features to provide a seamless user experience. Implement robust save and export functionalities to allow users to capture their designs dynamically and download them as HTML or JSON files for future use. Integrate backend services to handle essential features such as user authentication and paper storage, leveraging secure APIs

to facilitate data retrieval and management effectively. Conduct extensive testing across multiple devices and browsers to ensure the application is responsive, compatible, and performs well under various conditions, while addressing any bugs or errors that may arise. Once the application is thoroughly optimized, deploy it to a live server, gather feedback from users to understand their needs, and plan the integration of advanced features such as real-time collaboration, plugin support, and additional customization options to enhance functionality further.

To ensure the application's success, start by designing a user-friendly and intuitive interface that caters to both technical and non-technical users. The interface should include clear navigation, responsive design for compatibility across devices, and well-organized tools for editing and customization. Pay special attention to the accessibility standards to make the application usable for individuals with disabilities. Incorporate real-time visual feedback as users make changes, ensuring that edits are instantly reflected, which boosts user engagement and confidence in the platform.

Incorporate a robust backend infrastructure to support the core functionalities of the application. This includes implementing secure user authentication, enabling users to log in and manage their papers safely. For paper storage, consider integrating cloud services to allow scalability and reliability. Use version control mechanisms to let users save multiple versions of their designs, facilitating easy rollbacks if needed. Ensure data security with encryption techniques to protect sensitive user information and stored designs.

For post-deployment success, establish a feedback loop to continually improve the application based on user needs. Include an in-app feedback mechanism where users can share their suggestions or report issues directly. Regularly update the application to address bugs, incorporate new features, and adapt to emerging technologies or user demands. Additionally, provide detailed documentation and tutorials to help users understand and maximize the application's features. This proactive approach will help maintain user satisfaction, encourage adoption, and position the platform as a leader in its domain.

V. RESULT AND DISCUSSION

No Code Web-App Builder has successfully been delivering this weightless and responsive drag-and-drop website builder, along with live editing capabilities, customizing templates, managing components, and exportation of pure HTML/CSS/JS. Uses PHP and JSON for efficient saving/loading papers along with extensibility by means of plugin support. The system is a comprehensive and innovative platform designed to empower users without programming expertise to create and customize professional websites with ease. Built using a blend of modern technologies like HTML, CSS, JavaScript, Json, Bootstrap, and PHP, the paper combines versatility with user-friendliness. At its core, the platform features a robust library of pre-designed templates catering to diverse use cases, including portfolios, e-commerce websites,

blogs, and corporate sites. These templates serve as starting points, enabling users to quickly visualize their desired web structure and functionality. The drag-and-drop interface is one of the platform's most defining features, allowing users to effortlessly add, edit, move, and organize components such as text, images, buttons, forms, and navigation menus. The interface promotes hands-on interaction, making the process both intuitive and enjoyable for users.

Real-time previews elevate the customization experience by allowing users to see the impact of their changes instantly, eliminating guesswork and enhancing creative control. Advanced design flexibility is supported by Bootstrap, ensuring that all created websites are not only visually appealing but also fully responsive and optimized for various devices, from desktops to smartphones. On the backend, PHP plays a vital role in managing template storage, user sessions, and overall system operations, guaranteeing the platform's reliability and scalability for growing user bases. Yarn is utilized for efficient dependency management, streamlining the development and maintenance processes.

This paper addresses significant barriers faced by individuals and small businesses who lack the technical expertise or resources to develop websites from scratch. By simplifying the website creation process, the platform makes professional-grade web development accessible to a wider audience. Users can explore advanced customization options without the need for coding, enabling them to craft unique, tailored websites that align with their specific needs and branding requirements. Moreover, the paper fosters creativity and productivity, providing an intuitive space for users to experiment with design ideas and features.



Fig. 2. Home screen user interface

Figure 2 showcases the user interface of the home screen within the web application development environment. It highlights the component-based structure and available customization options.



Fig. 3. Sample web page components

Figure 3 presents a sample web page constructed using the platform's drag-and-drop interface. It demonstrates the integration of various components, including text, images, navigation, and a form, to create a functional webpage layout.

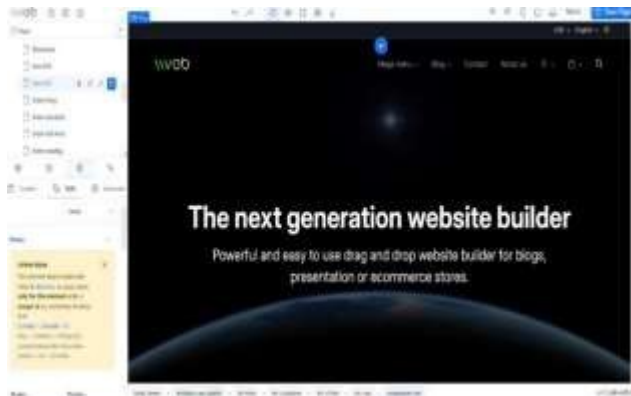


Fig. 4. Home page

The homepage presents an attractive introduction to a website builder. The user interface appears clean and informative, hinting at the ease of use mentioned in the text.



Fig. 5. CSS demonstration

The image displays a website builder interface with a design in progress. The page features a green background, a circular element, and a section with team member cards.

VI. CONCLUSION AND FUTURE WORK

The No-Code Web Application Builder paper is a great achievement towards democratization-it empowers all people

regardless of their experience to build fully operational, eye-catching websites. It established a drag-and-drop interface, prebuilt components, and design responsiveness that all effectively shorten barriers to traditional development complexities. The evidence for such impacts indicated the diversity of use cases as well as the capability of this service from small business sites to personal portfolios and an exuberant expression by the users concerning its ease of use and functionality. Comments from users further accentuate what users perceive as platform advantages in terms of being easy and fast to prototype as well as the possible areas of improvement such as having more templates, customization options, and integration features.

The system will also help in proving the importance of no-code solutions in closing the gap between technology and access for non-tech users altogether. Anyone can now efficiently control their presence on the web by empowering him or her to break the walls that inhibit access to web development. The paper would be paving the way for future improvements-having AI-powered recommendations for designs, making the template library huge, and enabling complicated workflows for advanced users.

The system not only witnesses the feasibility and possible future of no-code platforms but also strengthens their role in making web development more inclusive, effective, and accessible. An essential milestone in modern software development paradigms that will open new doors for future innovations and broader technology adoption can be found in this achievement.

REFERENCES

- [1] S. Käss, S. Strahinger and M. Westner, "A Multiple Mini Case Study on the Adoption of Low Code Development Platforms in Work Systems," in *IEEE Access*, vol. 11, pp. 118762-118786, 2023.
- [2] F. H. Khan, A. de Silva, G. Dusek, J. Davis, and A. Pang, "SmartCS: Enabling the creation of machine learning-powered computer vision mobile apps for citizen science applications without coding," *Citizen Science: Theory and Practice*, vol. 9, no. 1, pp. 1-16, 2024.
- [3] K. Rokis and M. Kirikova, "Exploring low-code development: A comprehensive literature review," *Complex Systems Informatics and Modeling Quarterly*, vol. 36, pp. 68-86, 2023.
- [4] E. Elshan, D. Germann, E. Dickhaut, and M. Li, "Faster, Cheaper, Better? Analyzing How Low Code Development Platforms Drive Bottom-Up Innovation," 2023.
- [5] J. SAE University College, "CIE214.1 NoCode Analysis Example 1 [Student Exemplar, SAE University College]," Creo, 2023.
- [6] D. Di Ruscio, D. Kolovos, J. de Lara, et al., "Low-code development and model-driven engineering: Two sides of the same coin?" *Softw. Syst. Model.*, vol. 21, pp. 437-446, 2022.
- [7] S. Liu, H. La, A. Willms, and R. A. Rhodes, "A 'No-Code' app design platform for mobile health research: Development and usability study," *JMIR Form Res.*, vol. 6, no. 8, pp. e38737, 2022.
- [8] S. Liu, H. La, A. Willms, and R. A. Rhodes, "A 'No-Code' app design platform for mobile health research: Development and usability study," *JMIR Form Res.*, vol. 6, no. 8, pp. e38737, 2022.
- [9] M. Moskal, "No-code application development on the example of Logotec App Studio platform," *Informatyka, Automatyka, Pomiar w Gospodarce i Ochronie Środowiska*, vol. 11, pp. 54-57, 2021.
- [10] S. S. Bhattacharyya and S. Kumar, "Study of deployment of 'low code no code' applications toward improving digitization of supply chain management," *J. Sci. Technol. Policy Manage.*, vol. 14, no. 2, pp. 271-287, Mar. 2023.
- [11] S. Siddique, M. Naveed, A. Ali, I. Keshta, M. I. Satti, A. Irshad, Z. Alomari, O. C. Edo, and O. A. Diekola, "An effective framework to improve the managerial activities in global software development," *Nonlinear Eng.*, vol. 12, no. 1, pp. 1-13,

- [12] S. A. A. Naqvi, M. P. Zimmer, P. Syed, and P. Drews, "Understanding the socio-technical aspects of low-code adoption for software development," in *Proc. ECIS*, 2023, pp. 1–17.
- [13] M. A. Al Alamin, S. Malakar, G. Uddin, S. Afroz, T. B. Haider, and A. Iqbal, "An empirical study of developer discussions on low-code software development challenges," in *Proc. IEEE/ACM 18th Int. Conf. Mining Softw. Repositories (MSR)*, May 2021, pp. 46–57.
- [14] S. Käss, S. Strahringer, and M. Westner, "Drivers and inhibitors of low code development platform adoption," in *Proc. IEEE 24th Conf. Bus.Informat. (CBI)*, vol. 1, Jun. 2022, pp. 196–205.
- [15] Y. Luo, P. Liang, C. Wang, M. Shahin, and J. Zhan, "Characteristics and challenges of low-code development," in *Proc. 15th ACM / IEEE Int. Symp. Empirical Softw. Eng. Meas. (ESEM)*, Oct. 2021, pp. 1–11.